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AI Invoice Reader – An AI-Powered Invoice Processing and Intelligent Financial Assistant

The domain of the Project:

Generative AI, Artificial Intelligence, OCR, Natural Language
Processing, Retrieval-Augmented Generation (RAG), Document
Intelligence and Financial Automation
Generative AI

Team Mentors :

Mr. Sreekanth Subramanian

Team Members:

Ms. Shruti Dahivalikar

Period of the project:

July 2026 to September 2026



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Declaration

The project titled “AI Invoice Reader – An AI-Powered Invoice Processing and Intelligent Financial Assistant” has been mentored by Mr. Sreekanth Subramanian, organised by SURE Trust, from July 2026 to September 2026 for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Shruti Dahivalikar —

Mentor's Name:

Mr. Sreekanth Subramanian

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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Executive Summary

The AI Invoice Reader is an AI-powered document-processing application designed to automate the extraction, organization, analysis and understanding of information contained in invoices.

Invoices are commonly received in different formats, including digitally generated PDFs, scanned documents and image-based documents. Manually reading these invoices and entering information into spreadsheets or accounting systems can be time-consuming, repetitive and prone to human error. The objective of this project is to develop an intelligent system that can reduce manual effort by automatically processing invoices and making their information easily accessible.

The system combines document processing, OCR, Large Language Models (LLMs), structured data extraction, database storage, Retrieval-Augmented Generation (RAG), financial analysis and an interactive Streamlit interface.

The application accepts invoice documents as input and processes them through an automated pipeline. For digitally generated PDFs, the system attempts to extract the available text directly. For scanned or image-based invoices, EasyOCR is used as a fallback mechanism to recognize text from the document. The extracted content is then passed to an AI-based invoice extraction component that converts unstructured invoice information into structured fields.

The extracted information can include details such as:

- Vendor or supplier name
- Invoice number
- Invoice date
- Due date
- Customer information
- Line items
- Quantities
- Prices
- Taxes
- Discounts
- Subtotal
- Total amount
- Currency
- Other relevant invoice information

The extracted invoice information is stored so that it can be retrieved and analyzed later. The application provides dashboard-based views for understanding invoice information and financial summaries.



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An AI Assistant is also included in the system. It allows users to ask natural-language questions about the available invoices. The RAG-based architecture retrieves relevant invoice information before generating an answer, helping the AI Assistant answer questions based on the user's invoice data rather than relying only on the model's general knowledge.

The project also includes an Admin Monitoring component for monitoring AI usage and API rate-limit information. This helps administrators understand AI request activity and monitor usage limitations.

The application is implemented using technologies including Python, Streamlit, EasyOCR, PyMuPDF, LangChain, LangGraph, Groq, Google Generative AI components, FAISS, Sentence Transformers, SQLite, Pandas and OpenPyXL.

The completed system demonstrates how Generative AI and document intelligence can be applied to a practical business problem by transforming unstructured invoices into structured, searchable and analyzable information.



Introduction

Background

Invoices are important business documents used for recording transactions between organizations, suppliers and customers. They contain important financial information such as invoice numbers, dates, products or services, quantities, taxes and total amounts.

Traditional invoice-processing workflows often involve manually reading invoices and entering information into accounting software or spreadsheets. This process becomes difficult when an organization receives a large number of invoices.

Invoices may also have different layouts and structures. Some invoices are digitally generated PDFs, while others may be scanned documents or photographs. This makes automated information extraction challenging because the location and formatting of information can vary from document to document.

Recent developments in Artificial Intelligence, Optical Character Recognition and Large Language Models provide an opportunity to automate this process.

The AI Invoice Reader project was developed to address this problem by creating an intelligent application capable of processing invoices, extracting important information and allowing users to interact with the extracted information through an AI Assistant.

Problem Statement

Manual invoice processing has several limitations:

- **Time consumption** – Employees may spend considerable time reading and entering invoice information.
- **Human errors** – Manual data entry can result in incorrect invoice numbers, amounts, dates or other fields.
- **Unstructured documents** – Different invoices use different layouts and formats.
- **Scanned documents** – Text may not be directly available in image-based PDFs.
- **Difficult information retrieval** – Finding specific information across many invoices can be inconvenient.
- **Limited analysis** – Raw invoice data does not automatically provide useful financial insights.
- **Scalability issues** – Manual processing becomes increasingly difficult as invoice volume increases.

Therefore, there is a need for an intelligent system that can automate invoice processing and provide users with an easier way to search, understand and analyze invoice information.



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Proposed Solution

The proposed solution is an AI-powered Invoice Reader that provides an end-to-end invoice processing workflow.

The system:

- Accepts invoice files from the user.
- Identifies and extracts available document text.
- Uses OCR for scanned/image-based invoices when necessary.
- Uses an LLM-based extraction process to identify structured invoice fields.
- Stores extracted information in a database.
- Provides dashboards for viewing invoice information.
- Provides financial summaries and analysis.
- Uses RAG to answer natural-language questions about invoices.
- Provides export functionality for extracted data.
- Provides an administrator monitoring interface for AI usage and rate-limit information.

Scope of the Project

The project covers the following major areas:

- Invoice document upload
- PDF processing
- OCR-based text extraction
- AI-based invoice information extraction
- Structured invoice storage
- Invoice search and retrieval
- Financial analysis
- Dashboard visualization
- Natural-language invoice Q&A
- Retrieval-Augmented Generation
- CSV and Excel export
- AI usage monitoring
- API rate-limit monitoring



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- Web-based deployment using Streamlit Community Cloud

Limitations

Although the system provides automated invoice processing, certain limitations remain.

- OCR accuracy can vary depending on document quality.
- Very low-quality or complex scanned documents may produce incorrect text.
- AI extraction depends on the quality and completeness of the source document.
- LLM APIs have token and request rate limits.
- AI-generated answers should be verified for critical financial decisions.
- The current application focuses primarily on invoice-related information rather than being a complete accounting system.
- Currency conversion and general financial questions may require additional capabilities beyond the invoice RAG system.

Innovation Component

The main innovation of the project is the combination of OCR + LLM-based information extraction + structured storage + RAG + financial analysis within a single application.

Instead of treating an invoice simply as a PDF, the system transforms it into structured information that can be searched, analyzed and queried using natural language.

The project also demonstrates the integration of multiple AI technologies into a practical business workflow rather than using an LLM as an isolated chatbot.



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Project Objectives

The major objectives of the AI Invoice Reader project are:

Objective 1 – Automate Invoice Processing

Develop an application that reduces manual effort involved in reading and processing invoices.

Objective 2 – Extract Structured Information

Automatically identify important invoice fields from unstructured invoice documents.

Objective 3 – Support Scanned Documents

Use OCR technology to process invoices where text cannot be extracted directly from the PDF.

Objective 4 – Improve Information Retrieval

Store invoice information in a structured form so that users can easily access previously processed invoices.

Objective 5 – Implement Intelligent Q&A

Allow users to ask natural-language questions about their invoice data.

Objective 6 – Implement RAG

Use Retrieval-Augmented Generation so that AI responses are grounded in the available invoice information.

Objective 7 – Provide Financial Insights

Generate useful summaries and financial information from processed invoices.

Objective 8 – Enable Data Export

Allow users to export invoice information into common formats such as CSV and Excel.

Objective 9 – Monitor AI Usage

Provide administrators with information about AI request usage and API rate limits.

Objective 10 – Build a User-Friendly Interface

Develop an interactive web application using Streamlit so that users can access the functionality without requiring programming knowledge.



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Methodology & Results

The SURE Trust template specifically asks for the methods/technology, tools/software, data collection approach, architecture, screenshots and GitHub link.

Methodology

The project follows a modular pipeline consisting of several stages.

Stage 1 – Invoice Upload

The user uploads an invoice document through the Streamlit interface.

The application accepts invoice documents and initiates the processing workflow.

Stage 2 – Document Processing

The document processor examines the invoice and attempts to obtain its text.

For digitally generated PDFs, text can be extracted directly from the document.

For scanned invoices where normal text extraction is insufficient, the system uses OCR.

Stage 3 – OCR Processing

EasyOCR is used to recognize text from scanned or image-based invoice pages.

The PDF pages can be rendered into images using PyMuPDF, after which image processing and OCR are performed.

This approach allows the system to process both digitally generated and scanned invoices.

Stage 4 – AI-Based Information Extraction

The extracted invoice text is passed to the AI extraction component.

A Large Language Model is used to identify relevant invoice fields and convert unstructured text into structured invoice information.

The structured output is validated using data models.

Stage 5 – Database Storage

The extracted invoice information is stored in the application's database.

The database allows processed invoices and AI usage information to be maintained for future retrieval.

Stage 6 – Dashboard Generation

The application presents processed invoice information through an interactive dashboard.

Users can view invoice details and financial information in a structured format.



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Stage 7 – Financial Analysis

The system processes invoice information to provide financial summaries and insights.

This can help users understand invoice totals and other financial characteristics without manually calculating them.

Stage 8 – RAG-Based AI Assistant

The AI Assistant allows users to ask questions about their invoices.

The RAG workflow retrieves relevant invoice information and provides it as context to the language model.

The model then generates a natural-language response based on the retrieved information.

Stage 9 – Export

Processed invoice information can be exported to formats such as CSV and Excel.

This makes the extracted information useful for further analysis and reporting.

Stage 10 – Administration and Monitoring

An Admin Monitoring section provides information related to AI usage, recent AI requests, Groq rate-limit information, and remaining request/token information where available.

This is particularly useful because cloud-based AI models operate under usage limits.

Technologies Used

Technology	Purpose
Python	Core programming language
Streamlit	Web application interface
EasyOCR	OCR for scanned invoices
PyMuPDF	PDF reading and page rendering
LangChain	LLM and AI application framework
LangGraph	AI workflow/agent orchestration
Groq	LLM inference



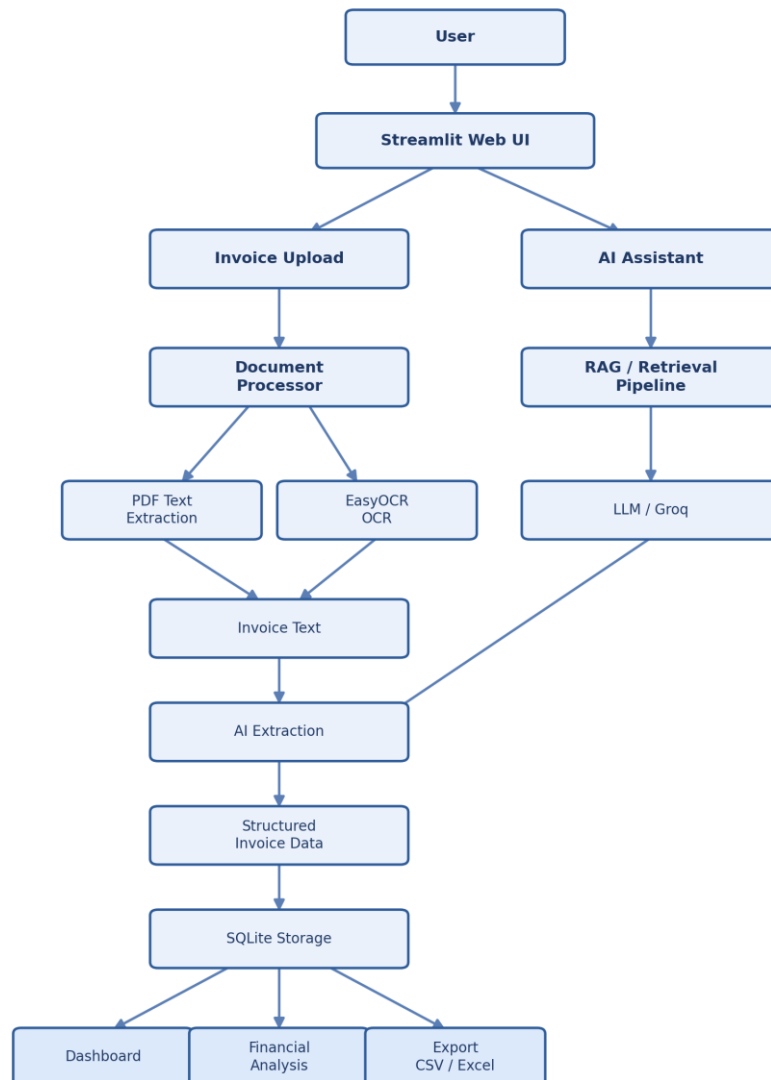
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Google Generative AI	Generative AI integration
Sentence Transformers	Text embeddings
FAISS	Vector similarity search
SQLite	Local structured data storage
Pandas	Data processing and analysis
OpenPyXL	Excel export
Plotly	Data visualization
Pydantic	Structured data validation
Python-dotenv	Environment variable management
Git/GitHub	Version control and repository management



Project Architecture

The application follows a modular architecture.



Major Modules

Document Processing Module

Responsible for extracting text from invoice documents. It supports PDF text extraction, page rendering, OCR fallback and image preprocessing.



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AI Extraction Module

Responsible for converting invoice text into structured invoice information. It uses an LLM and structured output validation.

Storage Module

Responsible for storing processed invoice information and application-related data.

Financial Service

Responsible for calculating and presenting financial information derived from invoices.

Invoice Service

Acts as a service layer connecting document processing, AI extraction and invoice storage.

RAG Module

Responsible for retrieving relevant invoice information before generating AI responses.

AI Agent

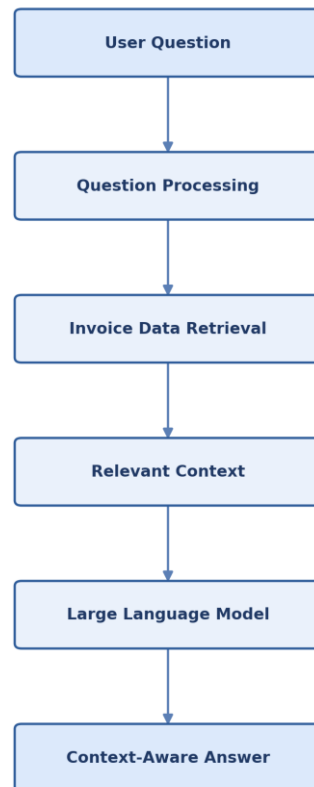
The application includes an AI agent workflow for interacting with invoice information and generating responses.

Admin Dashboard

The admin section provides monitoring information such as AI usage, recent requests and rate-limit information.

RAG Architecture

The Retrieval-Augmented Generation system can be represented as:



The advantage of this architecture is that the model can use the available invoice information as context when generating an answer.

For example, a user can ask “What is the total amount of the invoice?” or “Which vendor issued this invoice?”

The system attempts to retrieve relevant invoice information and use it to generate the response.

Results

The project successfully demonstrates the following capabilities:

- **Invoice Upload** – Users can upload invoice documents through the web interface.
- **Text Extraction** – The system can extract text from supported PDF documents.
- **OCR** – Scanned/image-based invoices can be processed using EasyOCR.
- **Structured Extraction** – Invoice information can be converted into structured fields.
- **Invoice Dashboard** – Users can view processed invoice information through the Streamlit interface.
- **Financial Overview** – The application provides financial information derived from processed invoices.



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- **AI Assistant** – Users can interact with invoice information through natural-language questions.
- **Data Export** – Invoice information can be exported for external analysis.
- **Admin Monitoring** – Administrators can monitor AI usage and rate-limit information.
- **Cloud Deployment** – The application has been configured for deployment using Streamlit Community Cloud.

Screenshots

Figure 1 – Home / Invoice Upload

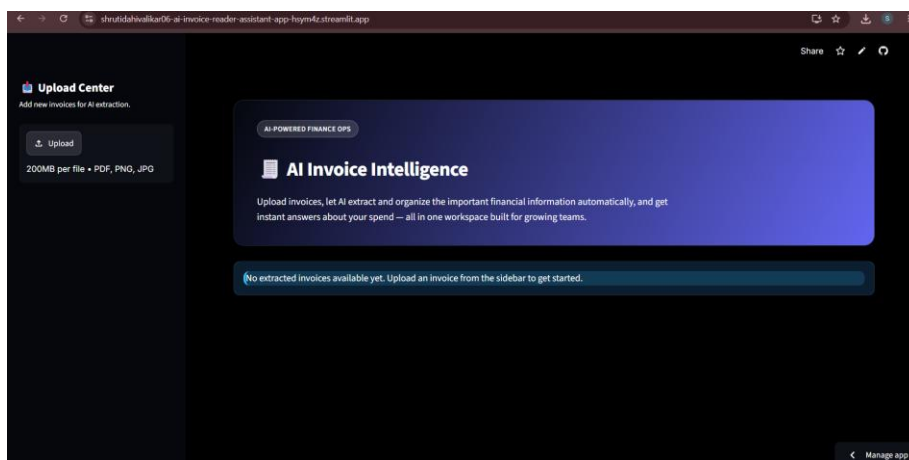


Figure 2 – Processed Invoice

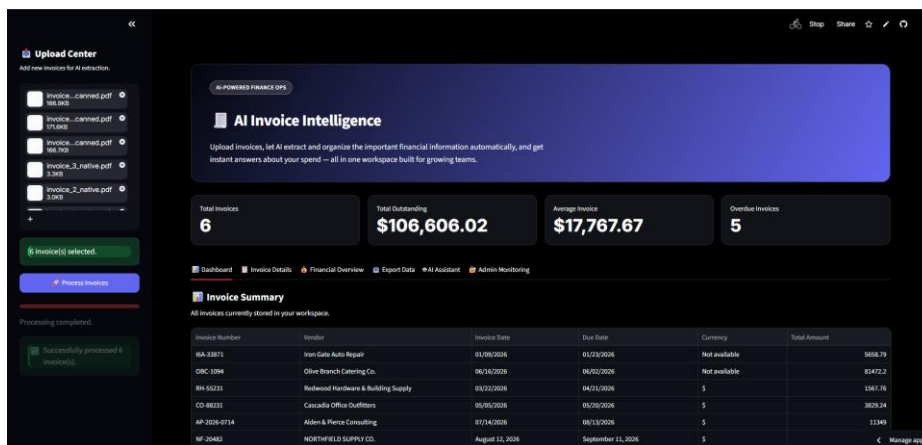


Figure 3 – Dashboard



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Dashboard Invoice Details Financial Overview Export Data AI Assistant Admin Monitoring

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Invoice Summary

All invoices currently stored in your workspace.

Invoice Number	Vendor	Invoice Date	Due Date	Currency	Total Amount
16A-33871	Iron Gate Auto Repair	01/09/2026	01/23/2026	Not available	5658.79
OBC-1094	Olive Branch Catering Co.	06/02/2026	06/16/2026	Not available	81472.2
RH-55231	Redwood Hardware & Building Supply	03/22/2026	04/21/2026	\$	1567.76
CO-88231	Cascadia Office Outfitters	05/05/2026	05/20/2026	\$	3829.24
AP-2026-0714	Alden & Pierce Consulting	07/14/2026	08/13/2026	\$	11349
NF-20482	NORTHFIELD SUPPLY CO.	August 12, 2026	September 11, 2026	\$	2729.03

Invoice Data Quality

16A-33871 — Iron Gate Auto Repair

- Missing currency

OBC-1094 — Olive Branch Catering Co.

- Missing currency

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Figure 4 – Financial Overview

Financial Overview

Total Invoices: 6

Total Amount: \$108,606.02

Total Amount: \$17,767.67

Overdue Invoices

16A-33871 Iron Gate Auto Repair | \$5,658.79

OBC-1094 Olive Branch Catering Co. | \$81,472.2

RH-55231 Redwood Hardware & Building Supply | \$1,567.76

AP-2026-0714 Alden & Pierce Consulting | \$11,349.00

CO-88231 Cascadia Office Outfitters | \$3,829.24

Invoices Due Soon

NF-20482 NORTHFIELD SUPPLY CO. | \$2,729.03 | Due in 3 days

Vendor Spending

Vendor	Invoice Count	Total Spending
Redwood Hardware & Building Supply	1	\$1,567.76
Iron Gate Auto Repair	1	\$5,658.79
Olive Branch Catering Co.	1	\$81,472.2
Alden & Pierce Consulting	1	\$11,349
Cascadia Office Outfitters	1	\$3,829.24

New Order: Olive Branch Catering Co. | Total Spending: \$81,472.2 | Invoices: 1

Monthly Spending

Month	Total Spending
August 2026	\$17,767.67
January 2026	\$5,658.79
June 2026	\$81,472.2
March 2026	\$1,567.76
July 2026	\$11,349
May 2026	\$3,829.24

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Figure 5 – AI Assistant

Dashboard Invoice Details Financial Overview Export Data AI Assistant Admin Monitoring

Monthly AI Summary

Select month

2026-08

Generate Monthly AI Summary

Ask Your Invoices

Ask questions about your invoices using AI-powered semantic search and analysis.

e.g. What is the total amount of the new invoice?

Clear

Ask AI

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Figure 6 – Admin Monitoring



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Dashboard Invoice Details Financial Overview Export Data AI Assistant Admin Monitoring

Admin Monitoring

Admin access required to view AI usage and system monitoring.

Admin Username

admin

Admin Password

Press Enter to apply

Login as Admin

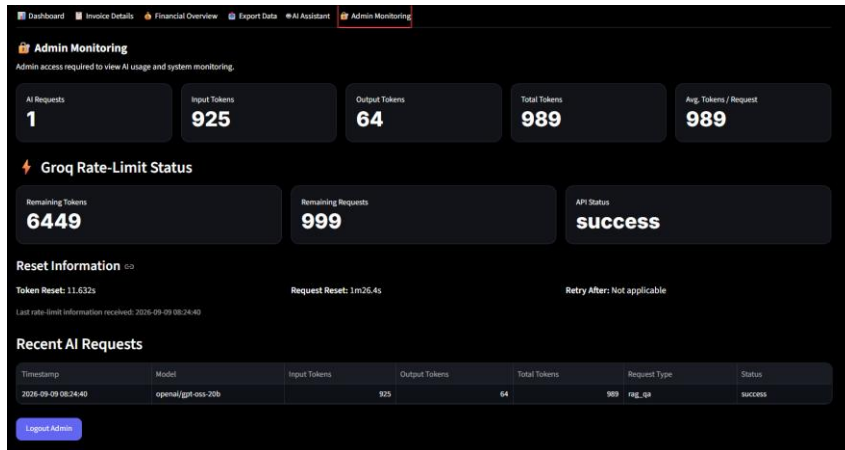
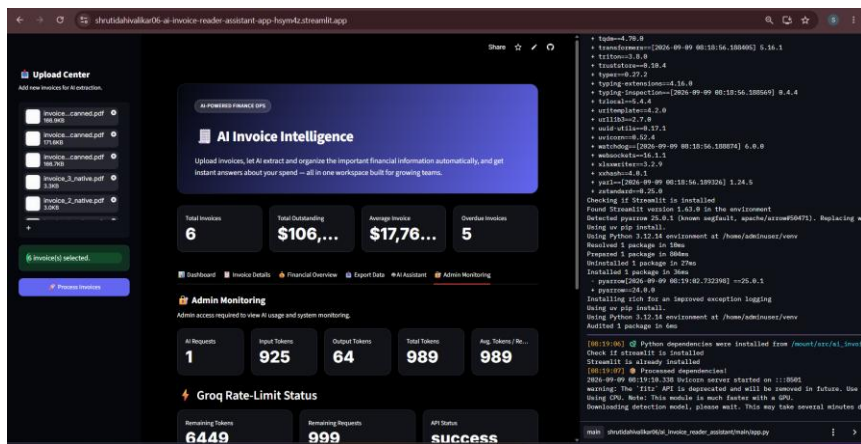


Figure 7 – Successful Deployment



Project GitHub Link

GitHub Repository: AI Invoice Reader Assistant – <https://github.com/sure-trust/DAHIVALIKAR-SHRUTI-SURYAKANT-g6-gen-ai/tree/main/Final%20capstone%20project/AI-Invoice-Reader>

The repository contains the project source code and supporting files.



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Social / Industry relevance of the project

The AI Invoice Reader has practical relevance for organizations that process large numbers of invoices.

Reduction in Manual Work

Automating invoice extraction can reduce repetitive manual data-entry tasks.

Employees can spend less time manually reading documents and more time on activities that require human judgment.

Improved Productivity

An automated workflow can process invoice information faster than a completely manual workflow.

Reduced Data Entry Errors

Structured AI extraction can help reduce errors associated with manually copying information from invoices.

However, important financial information should still be validated before being used for high-impact decisions.

Better Accessibility of Financial Information

The AI Assistant provides a natural-language interface for interacting with invoice information.

Users do not necessarily need to manually search through every document to locate information.

Small Business Applications

Small businesses may benefit from automated invoice processing because they often have limited administrative resources.

Accounting and Finance Applications

The technology can potentially be integrated into:

- Accounting workflows
- Accounts payable systems
- Procurement systems
- Expense management
- Financial reporting



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- Vendor management

Rural and Emerging Workforce Relevance

The project demonstrates how modern AI technologies can be applied to practical business problems and provides hands-on exposure to technologies that are increasingly relevant to the modern software industry.



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Learning & Reflection

The SURE Trust template asks that new technical and management learning and the overall project experience be documented.

Technical Learning

During the development of the AI Invoice Reader, significant learning was gained in the following areas.

Python Development

The project strengthened practical Python programming skills through development of a complete multi-module application.

Streamlit

The project provided practical experience in developing interactive web applications using Streamlit.

OCR

Working with EasyOCR provided experience in processing scanned documents and understanding the challenges associated with extracting information from images.

PDF Processing

The project provided practical experience with PDF text extraction and page rendering using PyMuPDF.

Generative AI

The project provided hands-on experience with Large Language Models and their application to structured information extraction.

LangChain

Experience was gained in using LangChain components for building AI-powered applications.

LangGraph

The project introduced workflow and agent orchestration concepts using LangGraph.

RAG

The project provided practical understanding of Retrieval-Augmented Generation and how retrieved context can be used to improve the relevance of LLM responses.

Vector Search

FAISS and embeddings provided experience with semantic retrieval and vector-based information search.



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Database Management

SQLite was used to store structured invoice and application information.

API Integration

The project provided experience integrating cloud-based AI APIs and handling API-related errors and rate limits.

Environment Variables and Security

Sensitive configuration such as API keys and administrator credentials were maintained through environment variables/secrets rather than hard-coded into the application.

Git and GitHub

The project improved understanding of:

- Git commits
- Branches
- Remote repositories
- Repository organization
- Deployment workflows
- Managing project versions

Cloud Deployment

The project provided practical experience deploying a Python/Streamlit application through Streamlit Community Cloud and troubleshooting dependency and deployment issues.

Deployed link:- <https://shrutidahivalikar06-ai-invoice-reader-assistant-app-hsym4z.streamlit.app/>

Problem-Solving Experience

One of the most important learnings from the project was understanding that developing an AI application involves more than simply connecting an LLM API.

Several practical challenges were encountered, including:

- Dependency management
- PDF processing
- OCR reliability
- Different invoice formats
- LLM API rate limits



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- Deployment configuration
- Environment variables
- RAG response quality

These challenges helped develop a systematic debugging approach.

API Rate-Limit Learning

The project also provided practical experience with AI API token limits.

During testing, a Groq API request returned a 429 rate-limit error because the request would have exceeded the model's tokens-per-minute limit.

This highlighted the importance of:

- Monitoring token usage
- Handling API errors
- Implementing retry strategies
- Monitoring rate-limit headers
- Designing efficient prompts
- Avoiding unnecessary AI calls

This experience is particularly valuable when building production-oriented AI applications.

Overall Experience

The project provided an opportunity to work through the complete lifecycle of an AI application:

Idea → Design → Development → Testing → Debugging → GitHub → Deployment → Monitoring

The project demonstrated that successful AI application development requires a combination of programming, AI knowledge, system design, debugging, data handling, deployment and problem-solving skills.



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Conclusion and Future Scope

Conclusion

The AI Invoice Reader successfully demonstrates the application of Artificial Intelligence and Generative AI to automate invoice processing.

The project combines OCR, PDF processing, Large Language Models, structured data extraction, database storage, RAG, financial analysis and a user-friendly web interface into a single application.

The system provides an automated workflow for processing invoices and converting unstructured documents into structured and searchable information.

The AI Assistant further improves accessibility by allowing users to interact with invoice data using natural language.

The project also demonstrates important real-world engineering concepts such as API integration, rate-limit management, environment-variable security, modular application design, version control and cloud deployment.

Overall, the project demonstrates that Generative AI can be used not only for conversational applications but also for practical document-processing and business-automation use cases.

Future Scope

The AI Invoice Reader can be extended significantly in future versions.

Multi-Format Document Support

Future versions could support:

- JPG
- PNG
- TIFF
- DOCX
- Email attachments
- Other business documents

Improved OCR

The OCR pipeline could be enhanced using:

- Better image preprocessing
- Deskewing
- Noise removal



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- Resolution enhancement
- Layout-aware OCR
- Specialized document AI models

This could improve extraction accuracy for low-quality invoices.

Multi-Language Invoice Support

The system could be extended to process invoices written in multiple languages.

This would make the application more useful for international organizations.

Advanced Financial Analytics

Future versions could provide:

- Monthly spending trends
- Vendor-wise spending
- Tax analysis
- Category-wise spending
- Expense forecasting
- Budget comparison
- Duplicate invoice detection
- Unusual invoice detection

Automated Currency Conversion

The application could integrate a live exchange-rate service so that invoices in different currencies can be converted into a selected base currency. For example:

- USD → INR
- EUR → INR
- GBP → INR

Improved AI Assistant

The AI Assistant could be extended to answer broader financial questions, such as:

- “How much did we spend this month?”
- “Which vendor has the highest total?”
- “Compare this month’s invoices with last month.”
- “Which invoices are overdue?”



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- “Show invoices above ₹1 lakh.”

Accounting Software Integration

Future versions could integrate with systems such as:

- ERP software
- Accounting platforms
- Procurement systems
- Expense management platforms

This could allow extracted invoice data to flow automatically into existing business systems.

Authentication and Role-Based Access

Future versions could implement multiple user roles:

- Administrator
- Accountant
- Manager
- Employee
- Viewer

Each role could have different permissions.

Cloud Database

Instead of local SQLite storage, a production version could use a cloud database for:

- Scalability
- Multi-user access
- Backup
- Centralized data management

Better Rate-Limit Management

Future versions could implement:

- Automatic retry with exponential backoff
- Request queuing
- Token budgeting
- Prompt optimization
- Model fallback



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- Usage alerts
- Daily/monthly usage dashboards

This would make the application more robust for larger workloads.